

Biodiversity and Conservation | CBSE

Biology Class XII Notes

By Ramakanta Biswal February 15, 2020

➤ **BIODIVERSITY**

- ✓ The combined diversity at all the levels of biological organization.
- ✓ Popularised by: Edward Wilson

→ ***Types of Biodiversity***

- **Genetic Diversity**: High degree of diversity shown by a single species over a distributional range.
 - Example-1: *Rauwolfia vomitoria* – Grows in the Himalayan ranges. Shows variation in the potency and concentration of the reserpine (active chemical)
 - Example-2: Occurrence of more than 50,000 strains of rice in India that are genetically different.

- Example-3: Occurrence of 1000 varieties of mango in India that are genetically different.

≈ **Species Diversity:** Diversity of the organisms at the level of species.

- Example: Amphibian species diversity is higher in the western ghats than the eastern ghats.

≈ **Ecological Diversity:** It represents the diversity in the availability of various types of ecosystem in a given area.

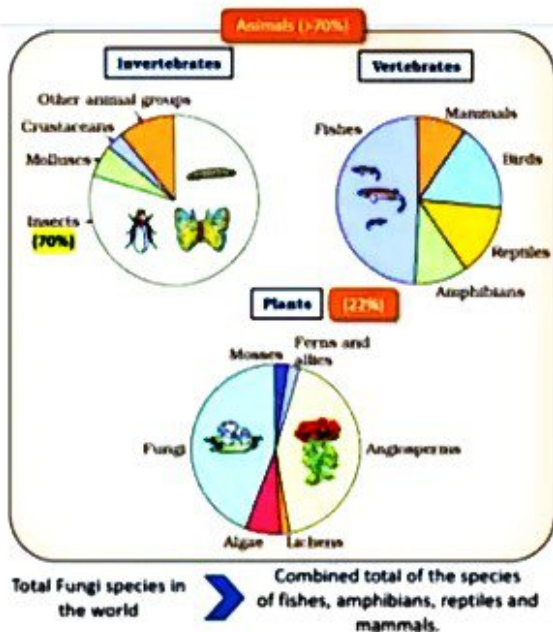
- Example: In comparison with a country like Norway, India has a greater ecosystem diversity with its deserts, rain forests, mangroves, coral reefs, wetlands, estuaries, and alpine meadows ecosystem.

→ ***Number of Species in India and on Earth***

- ✓ According to International Union for Conservation of Nature and Natural Resources (IUCN) (2004) data – total number of plant and animal species described so far is slightly more than 1.5 million.
- ✓ According to Robert May, the global species diversity is about 7 million.

→ ***Global Species Diversity :***

- ✓ No information on the number of prokaryotes.
- ✓ Biologists are not sure on the prokaryotic species.



→ India as one of the Mega Diversity countries of the world

- ✓ India – 2.4% of the world's land ; 8.1% of global species diversity
- ✓ 45,000 species of plants and twice as many animals have been recorded from India.

➤ PATTERNS OF BIODIVERSITY

→ Latitudinal Gradient:

- ✓ Species diversity decreases from equator to pole.
- ✓ Tropics have higher species than temperate/polar areas.
- ✓ Example: Colombia (near equator) – 1400 bird species ; India (Tropics)-1200 species; New York (Temperate)-105 species ; Greenland (polar)- 56 species.

☞ **Why does tropics have greater biological diversity than the temperate region?**



- ✓ The tropical latitudes have remained relatively undisturbed for millions of years and thus, had a long evolutionary time for species diversification.
- ✓ Tropical environments, unlike temperate ones, are less seasonal, relatively more constant and predictable. This promotes niche specialisation and leads to a greater species diversity.
- ✓ There is more solar energy available in the tropics, which contributes to higher productivity; this in turn might contribute indirectly to greater diversity.

➤ Species-Area relationships:

- ✓ It describes the relationship between the area of a habitat, or of part of a habitat, and the number of species found within that area.
- ✓ It is usually constructed for a single type of organism.
- ✓ Alexander von Humboldt observed that within a region species richness increased with increasing explored area, but only up to a limit.
- ✓ Species diversity is the number of diversity or the variety of species in a particular region. It can be explained by
 - Species Richness – The number of species present in a particular area.
 - Species even-ness – How evenly is the distribution of the number of each species.
- ✓ The relation between species richness and area for a wide variety of taxa (angiosperm plants, birds, bats, freshwater fishes) is a rectangular hyperbola.

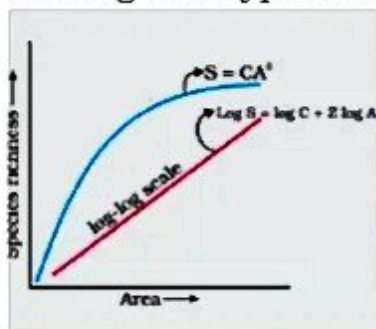


Fig 15.2 Showing species area relationships

- ✓ $\log S = \log C + Z \log A$

- ✓ The species area plots can give useful information such as the total number of species in a region and the rate of species increase with area between different regions (calculated from the slope of the curve).

➤ **Importance of Species Diversity:**

- ✓ Properties of a stable community:
 - should not show too much variation in productivity from year to year.
 - must be either resistant or resilient to occasional disturbances.
 - resistant to invasions by alien species.

→ ***David Tilman's long-term ecosystem experiments:***

- ✓ Experimented with outdoor plots.
- ✓ Plots with more species show less year-to-year variation in total biomass.
- ✓ Increased diversity contributes to higher productivity.

→ **Rivet Popper Hypothesis**

- ✓ Given by Paul Ehrlich
- ✓ Describes the impacts of extinction of the species on the ecosystem through an airplane analogy.
- ✓ In an airplane (ecosystem) all parts are joined together using thousands of rivets (species).
- ✓ If every passenger travelling in it starts removing a rivet to take home (causing a species to become extinct), it may not affect flight safety (proper functioning of the ecosystem) initially, but as more and more rivets are removed, the plane becomes dangerously weak over a period of time.
- ✓ Furthermore, which rivet is removed is also critical. For example, the loss of rivets on the wings (key species that drive major ecosystem functions) is obviously a more serious threat to flight safety than loss of a few rivets on the seats or windows inside the plane.

➤ **LOSS OF BIODIVERSITY**



- ✓ IUCN Red List: is the world's most comprehensive inventory of the global conservation status of biological species.
- ✓ The IUCN Red List (2004) documents the extinction of 784 species (including 338 vertebrates, 359 invertebrates and 87 plants) in the last 500 years.
- ✓ Examples of recent extinction:
 - Dodo
 - Quagga
 - Thylacine
 - Steller's Sea Cow
 - Bali, Javan and Caspian subspecies of Tiger.
- ✓ Effect of loss of biodiversity:
 - Decline in plant production
 - Lowered resistance to environmental perturbations such as drought
 - Increased variability in certain ecosystem processes such as plant productivity, water use and pest and disease cycles.

→ **The Evil Quartet of Causes of Loss of Biodiversity:**

1. Habitat loss and Fragmentation:

- ✓ When large habitats are broken up into small fragments due to various human activities, mammals and birds requiring large territories and certain animals with migratory habits are badly affected, leading to population declines.
- ✓ Degradation of many habitats by pollution also threatens survival of many species.
- ✓ Example-1: Tropical rain forests now cover only 6% of earth's land (reduced from 14%).
- ✓ Example-2: The Amazon rainforest being cleared for cultivation of soya beans or for conservation of grasslands for beef cattle.

2. Over-Exploitation:

- ✓ Over exploitation of natural resources.
- ✓ Many species have become extinct in the last 500 years. E.g. Passenger Pigeon
- ✓ Many marine fish species are overharvested, endangering the continued existence of the commercially important species.

3. Alien Species invasion:

- ✓ When an alien species (a new species that is not native to that habitat) is introduced it may become invasive and cause decline or

- ✓ Example-1: Nile Perch-introduced into Lake Victoria (East Africa) led to the extinction of more than 200 species of Cichlid fish in the lake.
- ✓ Example-2: Carrot grass (*Parthenium*); Lantana; Water hyacinth (*Eicchornia*).
- ✓ Example-3: African Catfish (*Clarias gariepinus*) introduction for aquaculture is a threat to the native catfishes in our rivers.

4. Co-Extinction:

- ✓ When a species becomes extinct, the plant and animal species associated with it in an obligatory (compulsory) manner also becomes extinct.
- ✓ Example-1: When a fish is extinct, the parasites associated in an obligatory manner also become extinct.
- ✓ Example-2: Co-evolved plant pollinator mutualism.

➤ **BIODIVERSITY CONSERVATION**

→ **Why Should We Conserve Biodiversity?**

1. **Narrowly Utilitarian Argument:**

- ✓ Views biodiversity as a source for various products
- ✓ Benefits from biodiversity:
 - Food (cereals, pulses, fruits)
 - Firewood
 - Fibre
 - Construction materials
 - Industrial products
 - Products of medical importance
- ✓ **Bioprospecting**: exploring molecular, genetic and species-level diversity for products of economic importance.

2. **Broadly Utilitarian Argument**

- ✓ Argues about many ecosystem services that nature provides.
- ✓ Example-1: Amazon forest produces 20% of total Oxygen in the atmosphere via photosynthesis.
- ✓ Example-2: Pollination – Ecosystem, provides pollinators layer – bees, bumblebees, birds and bats.
- ✓ Example-3: the aesthetic pleasures of walking through thick woods, watching spring flowers in full bloom or waking up to a bulbul's song in the morning.



3. Ethical Argument

- ✓ We share the planet with millions of plants, animals and microbe species.
- ✓ Every species has an intrinsic value, even if it does not have any economic value.
- ✓ We have a moral duty to care for their well-being and pass on our biological legacy in good order to future generations.

→ How Do We Conserve Biodiversity?

1. In situ conservation:

- ✓ Also known as **on site** conservation.
- ✓ Conservation and protection of the entire ecosystem – biodiversity at all levels is protected.
- ✓ Biodiversity Hotspots: Regions with very high levels of species richness and high degree of endemism, that are identified for maximum protection.
- ✓ Total biodiversity hotspots in the world = 34
- ✓ Biodiversity hotspots of India with exceptionally high biodiversity
 - Western Ghats & Sri Lanka:
 - Includes entire Western Ghats (and Sri Lanka)
 - Indo-Burma:
 - Includes entire North-eastern India, except Assam and Andaman group of Islands (and Myanmar, Thailand, Vietnam, Laos, Cambodia and southern China)
 - Himalaya:
 - Includes the entire Indian Himalayan region (and that falling in Pakistan, Tibet, Nepal, Bhutan, China and Myanmar)
- ✓ In India, ecologically unique and biodiversity-rich regions are legally protected as
 - biosphere reserves - 14
 - national parks - 90
 - sanctuaries - 448
- ✓ Sacred Groves:
 - These are forest fragments of varying sizes, which are communally protected, and which usually have a significant religious connotation for the protecting community.
 - Hunting and logging are usually strictly prohibited within these patches.

- Example-1: Khasi and Jaintia Hills in Meghalaya
- Example-2: Western Ghat of Karnataka and Maharashtra
- Example-3: Sarguja, Chanda and Bastar areas of Madhya Pradesh

2. Ex situ Conservation:

- ✓ Also known as **off site** conservation.
- ✓ Usually approached when the plant or animal is endangered or threatened and needs urgent measures to save from extinction.
- ✓ Here the threatened plants or animals are removed from their natural habitat and are placed in special protected areas where they can be given special care.
- ✓ Methods-1: Zoological park
 - many animals that have become extinct in the wild but continue to be maintained in zoological parks
- ✓ Method-2: Botanical garden
- ✓ Method-3: Wildlife Safari
- ✓ Method-4: Cryopreservation techniques
 - Using this technique, the gametes of threatened species can be preserved in viable and fertile condition for long periods.
 - The eggs can be fertilised using *in-vitro* technique.
- ✓ Method-5: Seed Bank
 - Using this method seeds of different genetic strains of commercially important plants can be kept for long periods.

→ Biodiversity Conservation-Measures at global level

The Earth Summit-1992

- ✓ Convention on Biological diversity held in Rio de Janeiro for conservation of biodiversity and sustainable utilisation of its benefits.

World Summit on Sustainable Development – 2002

- ✓ Held in Johannesburg, South Africa
- ✓ Commitment by 190 countries for a significant reduction in the current rate of biodiversity loss at global, regional and local levels.

